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Sseed Studio LoRa-E5 mini

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Sseed Studio LoRa-E5 mini

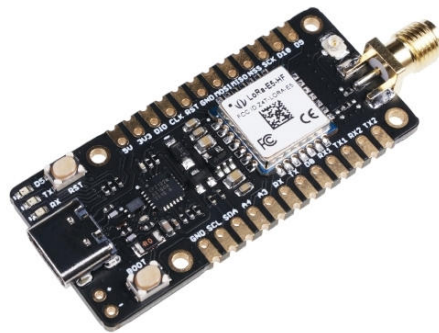
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Summarize

v2.6.99-cs1

Overview

LoRa-E5 mini is a compacted-sized development board suitable for the rapid testing and building of small-sized LoRa device, exposing all capabilities of Sseed Studio LoRa-E5 STM32WLE5JC module.



Hardware

The boards' LoRa-E5 Module packages a STM32WLE5JC SOC, a 32MHz TCXO, and a 32.768kHz crystal oscillator in a 28-pin SMD package. This STM32WLE5JC SOC is powered by ARM Cortex-M4 core and integrates Semtech SX126X LoRa IP to support (G)FSK, BPSK, (G)MSK, and LoRa modulations.

- LoRa-E5 STM32WLE5JC Module with STM32WLE5JC multiprotocol LPWAN single-core 32-bit microcontroller (Arm® Cortex® -M4 at 48 MHz) in 28-pin SMD package featuring:
 - Ultra-low-power MCU
 - RF transceiver (150 MHz to 960 MHz frequency range) supporting LoRa®, (G)FSK, (G)MSK, and BPSK modulations
 - 256-Kbyte Flash memory and 64-Kbyte SRAM
 - Hardware encryption AES256-bit and a True random number generator
- 1 user LED
- 2 serial communication (RX/TX) LEDs
- 1 boot/user and 1 reset push-button
- 32.768 kHz LSE crystal oscillator
- 32 MHz HSE oscillator
- Board connectors:
 - USB Type-C connector
 - +/- (battery) power input pins (3-5V)
 - SMA-K and IPEX antenna connectors
- Delivered with SMA antenna (per default IPEX connector is disconnected)
- Flexible power-supply options: USB Type C or 3-5V battery soldered to +/- pins
- Suitable for rapid prototyping of end nodes based on LoRaWAN, Sigfox, wM-Bus, and many other proprietary protocols
- All GPIOs led out from the LoRa-E5 STM32WLE5JC module
- 4x M2 mounting holes

More information about the board can be found at the [LoRa-E5 mini Wiki](#).

More information about LoRa-E5 STM32WLE5JC Module can be found here:

- [LoRa-E5 STM32WLE5JC Module Wiki](#)
- [LoRa-E5 STM32WLE5JC Module datasheet](#)
- [STM32WLE5JC datasheet](#)
- [STM32WLE5JC reference manual](#)
- [STM32WLE5JC on www.st.com](#)

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Supported Features

The Zephyr LoRa-E5 mini configuration supports the following hardware features:

Interface	Controller	Driver/Component
ADC	on-chip	adc
AES	on-chip	crypto
COUNTER	on-chip	rtc
CLOCK	on-chip	reset and clock control
FLASH	on-chip	flash
GPIO	on-chip	gpio
I2C	on-chip	i2c
MPU	on-chip	arch/arm
NVIC	on-chip	arch/arm
PINMUX	on-chip	pinmux
RADIO	on-chip	LoRa
SPI	on-chip	spl
UART	on-chip	serial port-polling; serial port-interrupt
WATCHDOG	on-chip	independent watchdog

Other hardware features are not yet supported on this Zephyr port.

The default configuration can be found in the defconfig and dts files:

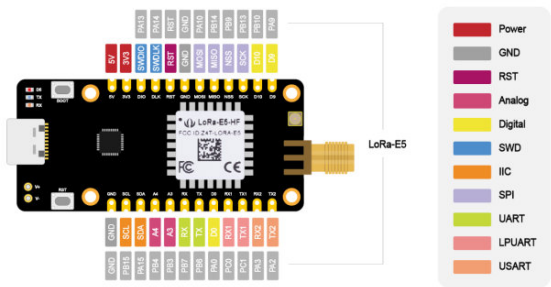
- boards/arm/lora_e5_mini/lora_e5_mini_defconfig
- boards/arm/lora_e5_mini/lora_e5_mini.dts

Connections and IOs

LoRa-E5 mini has 4 GPIO controllers. These controllers are responsible for pin muxing, input/output, pull-up, etc.

Available pins:

LoRa-E5 mini Pinout



Default Zephyr Peripheral Mapping:

- USART_1 TX : PB6
- USART_1 RX : PB7
- I2C_2_SCL : PB15
- I2C_2_SDA : PA15
- BOOT_PB : PB13
- LED_1 : PB5

System Clock

LoRa-E5 mini board System Clock could be driven by the low-power internal (MSI), High-speed internal (HSI) or High-speed external (HSE) oscillator, as well as main PLL clock. By default System clock is driven by the MSI clock at 48MHz.

Programming and Debugging

Applications for the `lora_e5_mini` board configuration can be built the usual way (see [Building an Application](#)).

In the factory the module is flashed with an DFU bootloader, an AT command firmware, and the read protection level 1 is enabled. So before you can program a Zephyr application to the module for the first time you have to reset the read protection to level 0. In case you use an st-link debugger you can use the STM32CubeProgrammer GUI to set the RDP option byte to `AA`, or use the STM32_Programmer_CLI passing the `--readunprotect` command to perform this read protection regression. The RDP level 1 to RDP level 0 regression will erase the factory programmed AT firmware, from which seed has neither released the source code nor a binary. Also, note that on the module the `BOOT0` pin of the SOC is not accessible, so the system bootloader will only be executed if configured in the option bytes.

Flashing

The LoRa-E5 mini does not include a on-board debug probe. But the module can be debugged by connecting an external debug probe to the 2.54mm header. Depending on the external probe used, `openocd`, the `stm32cubeprogrammer`, `pyocd`, `blackmagic`, or `jlink` runner can be used to flash the board. Additional notes:

- Pyocd: For STM32WL support Pyocd needs additional target information, which can be installed by

adding "pack" support with the following pyocd command:

[Explain this code](#)

```
$ pyocd pack --update
$ pyocd pack --install stm32w1
```

Flashing an application to LoRa-E5 mini

Connect the LoRa-E5 to your host computer using the external debug probe. Then build and flash an application. Here is an example for the [Hello World](#) application.

Run a serial host program to connect with your board: Per default the console on `usart1` is available on the USB Type C connector via the built-in USB to UART converter.

[Explain this code](#)

```
$ picocom --baud 115200 /dev/ttyACM0
```

Then build and flash the application.

[Explain this code](#)

```
# From the root of the zephyr repository
west build -b lora_e5_mini samples/hello_world
west flash
```

Debugging

You can debug an application in the usual way. Here is an example for the [Blinky](#) application.

[Explain this code](#)

```
# From the root of the zephyr repository
west build -b lora_e5_mini samples/basic/blinky
west debug
```

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